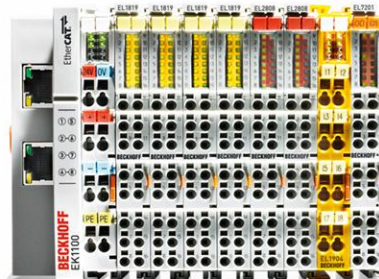


PLC_MOTION_LAYER

TwinCAT project

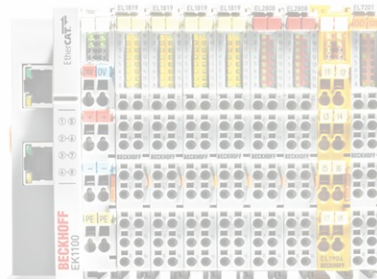
BECKHOFF



Daniel Hauer

Introduction to motion layer approach

- motivation for a motion layer
- use cases
- specific construction



Motivation:

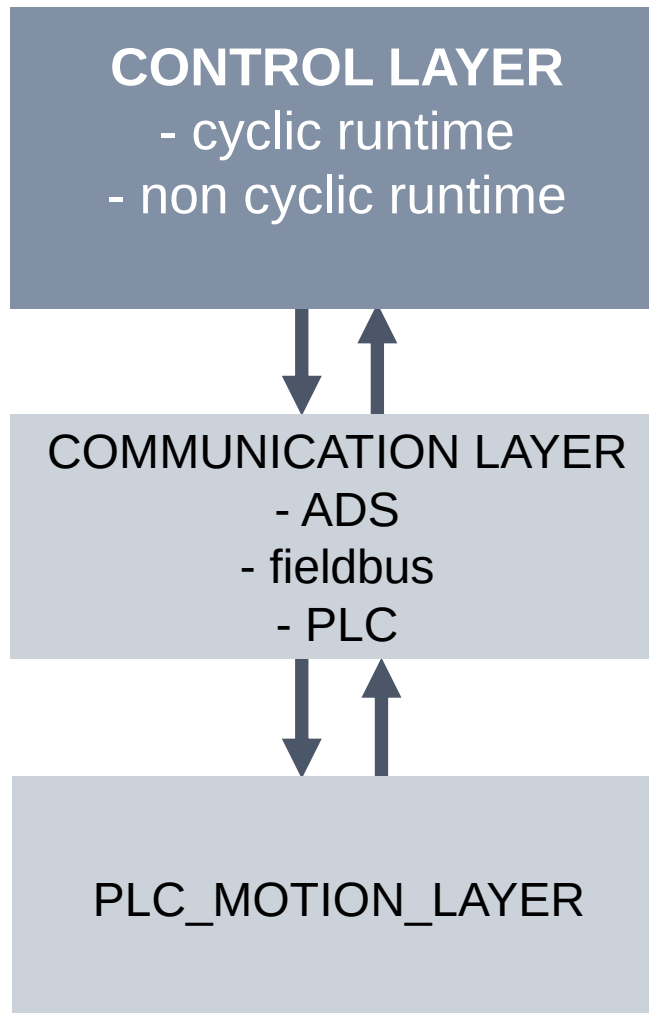
- **Centralized control:**
 - Sync communication
 - Async communication
 - Agnostic to communication technology
- **Decentralized execution of motion tasks**
 - Data consistency in cyclic multi task environments
 - Local machine with modular design specifications
 - Unified procedure for modular software architecture
 - Use of top layer consistent through different architectures

Motivation:

- **Use of TwinCAT supported/updated libraries**
 - Tc2_MC2
 - Tc2_MC2_Drive
 - Tc2_NC
 - Tc2_NCI
 - Tc2_PlcInterpolation
- **Open code base**
 - Migration to Tc3 MC in preparation
 - Customer/user specific changes possible
 - Access to code
 - Conversion to library possible by customer/user
- **Compiled PLC**
 - Source code is not on shipped machine

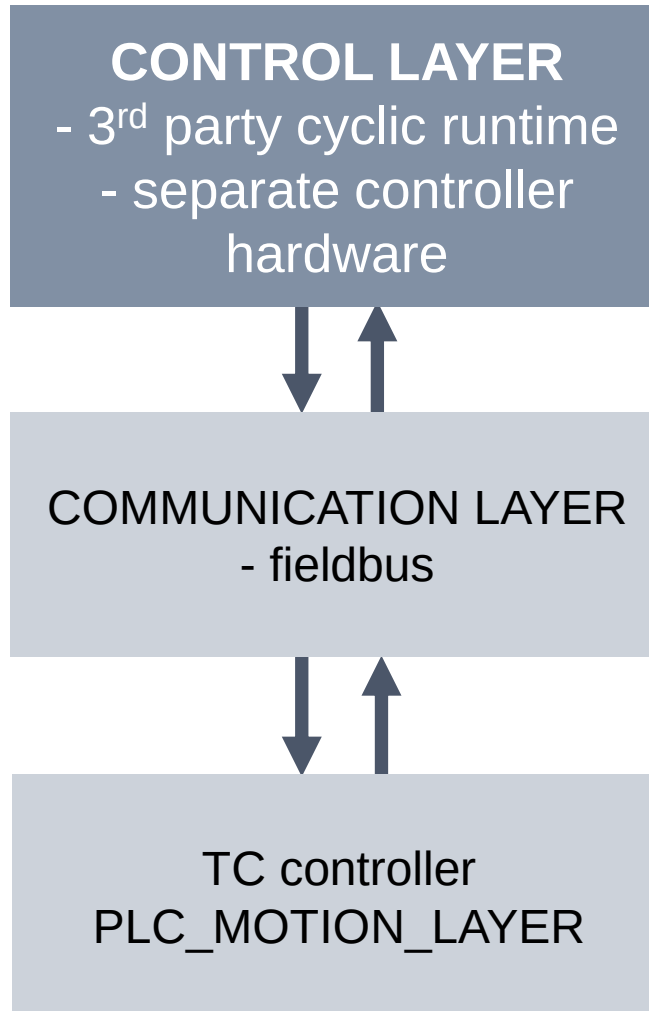
Motivation:

- Use of TwinCAT motion without detailed coding knowledge
- Transparency of communication layer
- Code base shall remain independent of control layer
- Configurable Options for specific libraries / TC functions
- Balanced load for configurable options in machine layout
- Stable cpu use for XFC applications



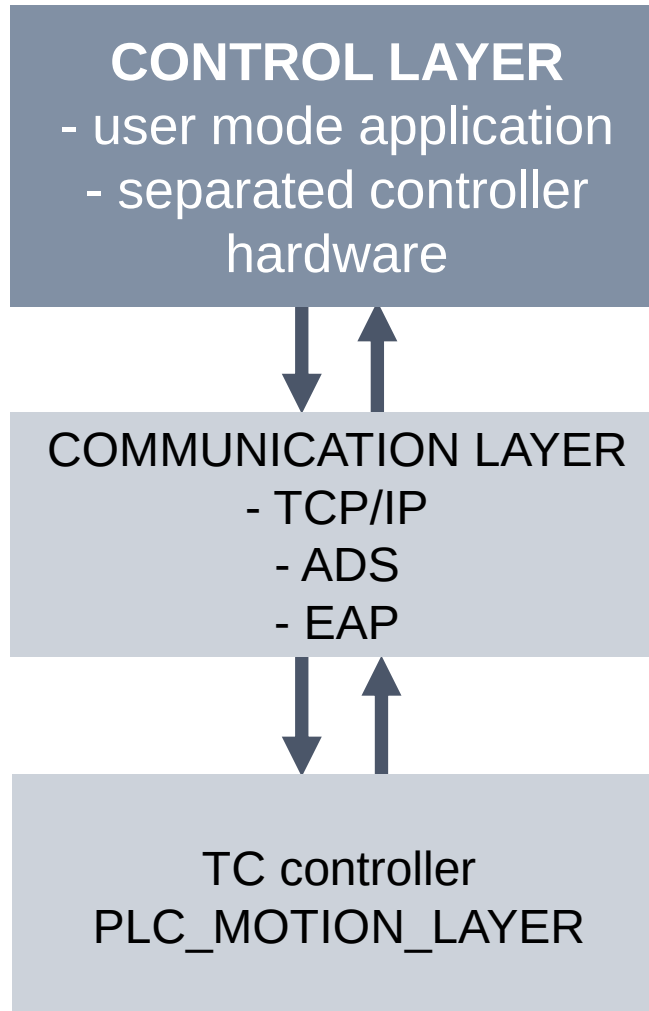
Use cases:

- Separate controller for machine logic
- Any fieldbus (EtherCAT, Profi...)
- Connected through TwinCAT mappings
- Execution of motion tasks in PLC_MOTION_LAYER TwinCAT controller



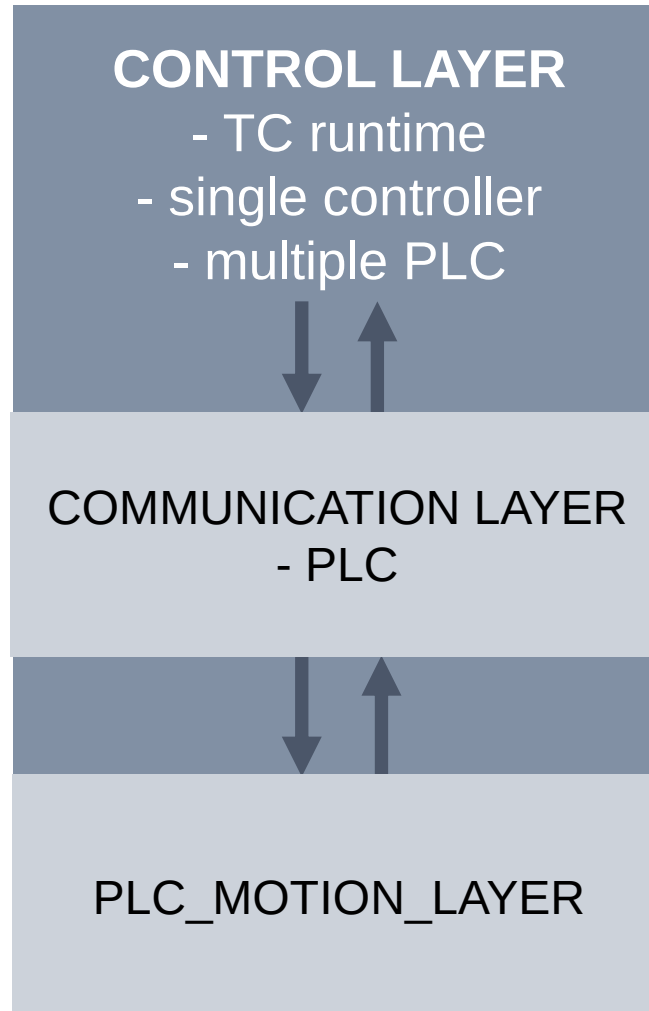
Use cases:

- Separate controller for machine logic
- Any network
- Connected through TwinCAT mappings
- Execution of motion tasks in PLC_MOTION_LAYER TwinCAT controller



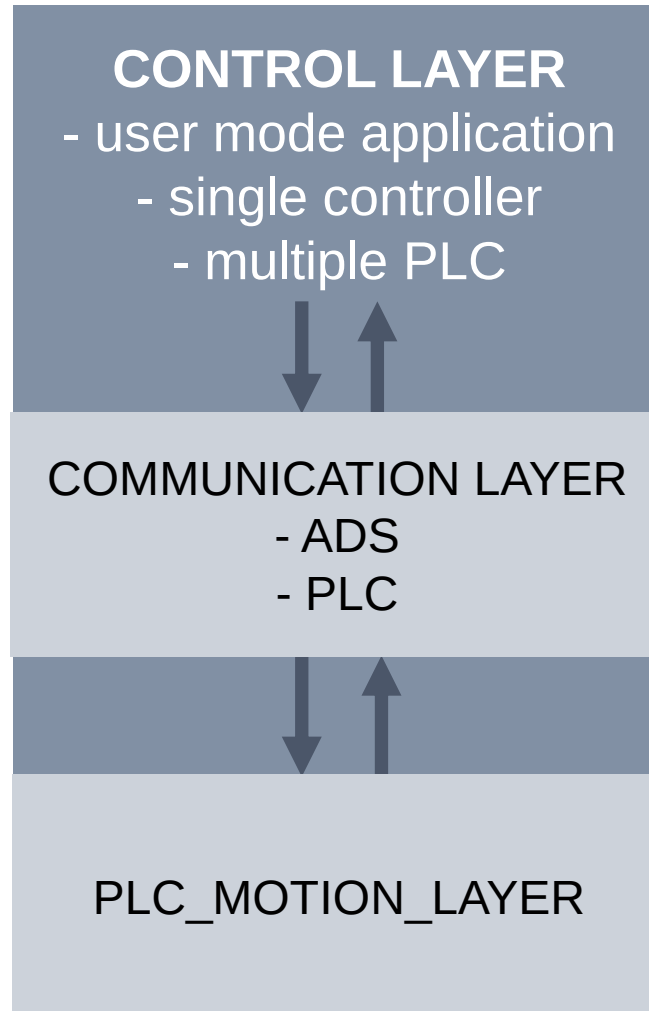
Use cases:

- One controller for machine logic
- Multiple PLC for machine logic
- Connected through TwinCAT mappings
- Execution of motion tasks



Use cases:

- One controller for machine logic
- User mode application AND/OR multiple PLC
- ADS for symbol access by user mode application
- TwinCAT mapping for connecting multiple PLCs for specific application purposes
- Execution of motion tasks



Specific construction:

- TwinCAT PLC project
- Use of specific syntactic code behaviour
- Software design
- Compiler defines / pragmas
- Logging system

TwinCAT project:

- Default TwinCAT project
 - Adjust core settings to target hardware
 - Add NC/PtP
 - Optional add NCI channel
- TwinCAT PLC
 - Add existing Item: PLC_MOTION_LAYER
 - Add task reference
 - Option: add compiler defines (NCI, CAM, BSD)
- Adjust constants in:
 - PLC_MOTION_LAYER/PLC_CONSTANT
- Compile
 - PLC_MOTION_LAYER Instance mapping is built

specific syntactic code behaviour:

- C like state machines
 - State changes **need not** consume one PLC cycle
- Same cycle response to command on cyclic interface
 - → since we're not using C (unfortunately, but ST in TwinCAT3 is almost as beautiful)
 - → cases have to be 'broken up'
 - → take a close look, it probably looks weird, but it is fast AND safe to use in combination with E_PROGRESS
 - → look at FB_MaAxisBase.MovePosBuffer try to do it with IF conditionals, then come back and take a second look ;-)

specific syntactic code behaviour:

- OnChange detection for new commands
 - Cyclic check whether the command has changed
- State always carries offset about progress of command (E_PROGRESS) (example follows in a few pages)
- Library FBs are called within states
 - FBs are called when required and not a cycle longer (same is true for all Ctrl-Wrappers)
 - Working with empty cyclic calls is the best way to build voodoo software, just don't do it!

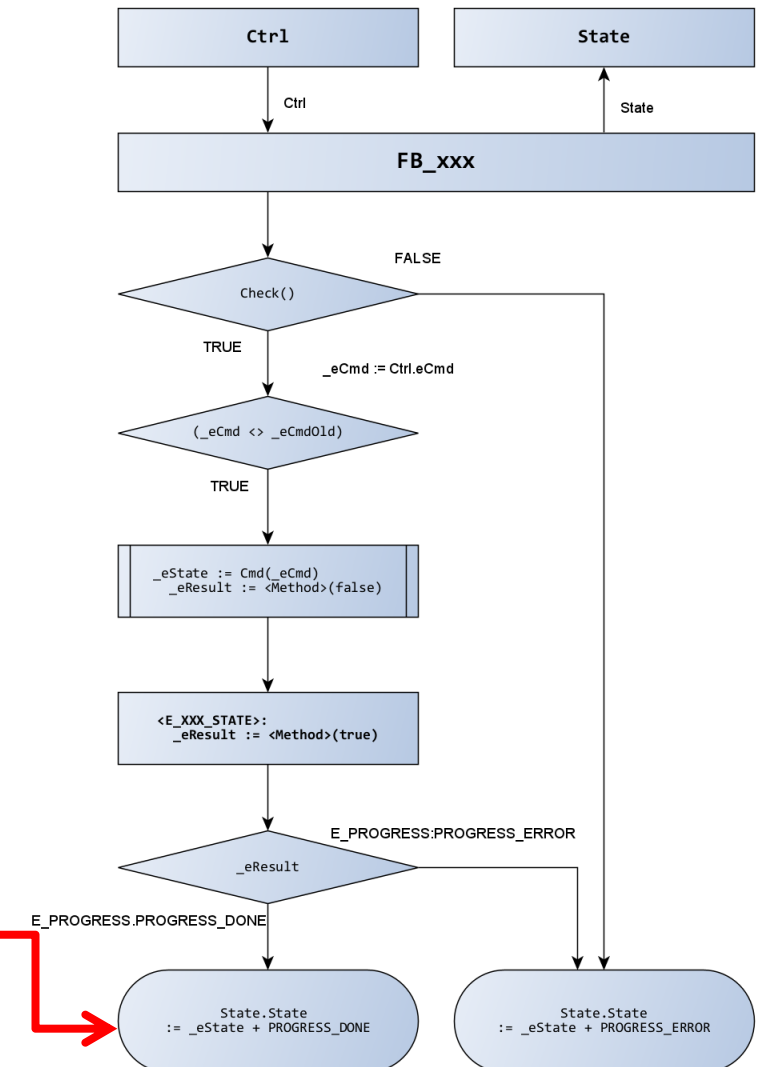
```
_eCmd := _Ctrl^[_nAxisIndex].eCmd;  
//-----  
//-----  
// OnChange react to command  
//-----  
IF (_eCmd <> _eCmdOld)  
THEN  
    // get busy  
    _eState := Cmd(_eCmd); // get execution state for command  
    _eCmdOld := _eCmd;  
    LogControl();  
END_IF
```

Software Design:

- Every TwinCAT function has dedicated wrapper
 - Separate namespaces
 - Optional library binding
- Cross communication via interfaces
 - NCI, CAM, XFC use interfaces in order to enable optional binding
 - If compiler define is not set, empty interfaces are used instead of instances
- Ctrl/State structures for commanding required function
 - PtP – ctrl/state
 - NCI – ctrl/state
 - CAMMING – ctrl/state
- Parameter structures carry required data for commanded function
 - PtP – (SetPos, SetVelo, SetAcc, MasterAxisIndex...)
 - NCI – (AxisGroupId, AxisIndex, MFunc, RParameter...)
 - CAMMING – (MasterAxisIndex, TableId...)

Software Design:

- State / Ctrl structures
 - Establishes unified access
 - Commands can simply be 'dropped' into Ctrl datafield
 - Enables asynchronous communication with PLC_MOTION_LAYER (e.g. C# via ADS, C/C++ via ADS, ADS over MQTT, ...)
 - Enables cyclic communication with PLC_MOTION_LAYER since structures can easily be mapped onto any cyclic fieldbus
TwinCAT supports (EtherCAT, Profinet, CanOpen, EAP, ...)
 - State is updated by PLC_MOTION_LAYER, so you can move on doing other stuff, come back and check completion/error.
- State feedback for cyclic class wrappers
 - **Always** combined with E_PROGRESS
 - You can filter your response by a simple modulo division
 - The result may be your entry point for your reaction to State.



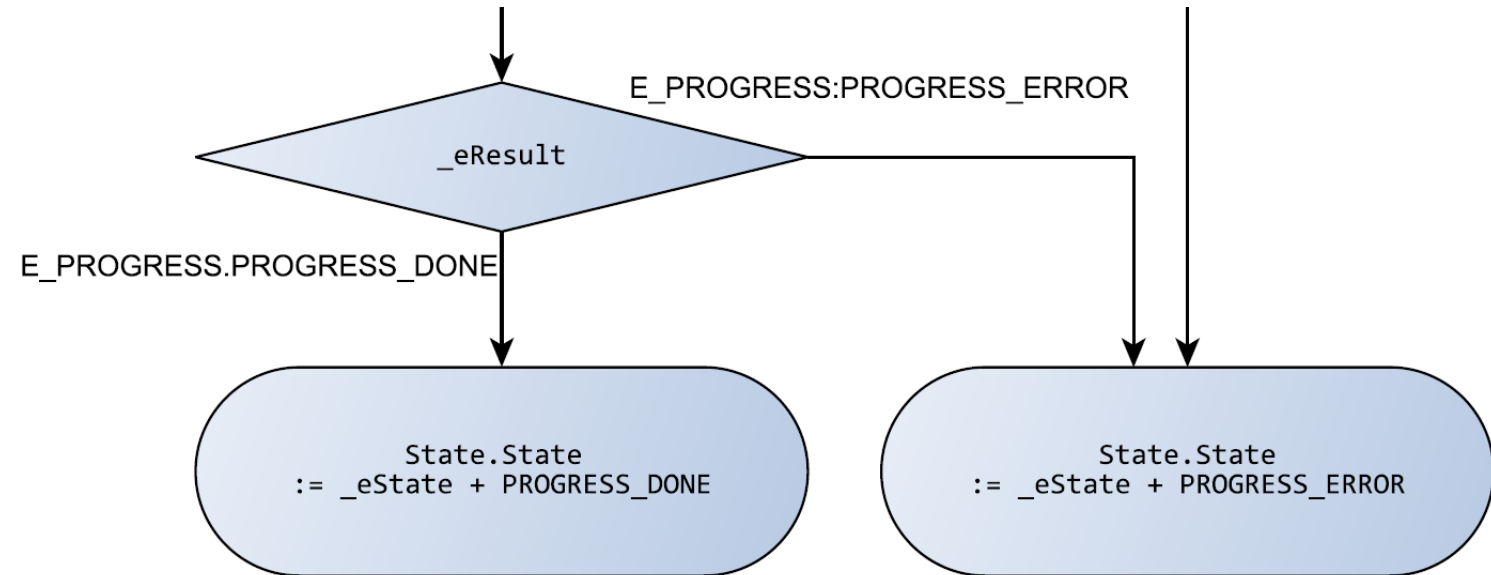
Software Design:

- E_PROGRESS
 - How far along is the command you just 'dropped'?
 - This enum shall help you to build an answer state machine and not just a simple **IF** conditional
 - This enum shall be used everywhere and I recommend you use it in your external control layer

```
E_PROGRESS  ▢ ×
1  {attribute 'qualified_only'}
2  //{attribute 'strict'}
3  {attribute 'to_string'}
4  TYPE E_PROGRESS :
5  (
6    // progress has 2 use cases
7    // 1. as offset to cyclic interface's state
8    //    for the requested command/function
9    //    e.g. State := <Enum equivalent to eCmd> + E_PROGRESS
10   //
11   // 2. as state feedback (result) from (any) method
12   PROGRESS_INVALID,
13   PROGRESS_NOT_EXIST      := 100,
14   PROGRESS_INIT           := 1000,
15   PROGRESS_BUSY           := 2000,
16   PROGRESS_PREPARE        := 3000,
17   PROGRESS_STARTUP        := 4000,
18   PROGRESS_CHECK          := 5000,
19   PROGRESS_OCCUPIED       := 6000,
20   PROGRESS_WORKING        := 7000,
21   PROGRESS_STILL_WORKING  := 8000,
22   PROGRESS_ERROR          := 9000,
23   PROGRESS_DONE           := 10000
24 )UINT;
25 END_TYPE
```


Software Design:

- E_PROGRESS (example)
 - How far along is the command you just 'dropped'?
 - This enum shall help you to build an answer state machine
 - This enum shall be used everywhere and I recommend you use it in your extern control layer too.



Software Design:

- E_PROGRESS (example)
 - How far along is the command you just 'dropped'?
 - See example skeleton →
 - This enum shall help you to build an answer state machine
- This enum shall be used everywhere and I recommend you use it in your extern control layer too.

```
//-----  
//-----  
// react to operation modes (how to do it in an easy self documenting way)  
//-----  
//-----  
CASE (GVL_AXIS.State[i].eState MOD E_PROGRESS.PROGRESS_DONE)  
OF  
    E_AXIS_STATE.AXIS_INIT:  
        ;// do something after success  
  
    E_AXIS_STATE.AXIS_MOVE_POS::  
        ;// do something after success  
ELSE  
    CASE (GVL_AXIS.State[i].eState MOD E_PROGRESS.PROGRESS_ERROR)  
    OF  
        E_AXIS_STATE.AXIS_INIT:  
            ;// do something after success  
  
        E_AXIS_STATE.AXIS_MOVE_POS::  
            ;// do something after success  
    ELSE  
        ;// down the rabbit hole you go from here  
    END_CASE  
END_CASE
```

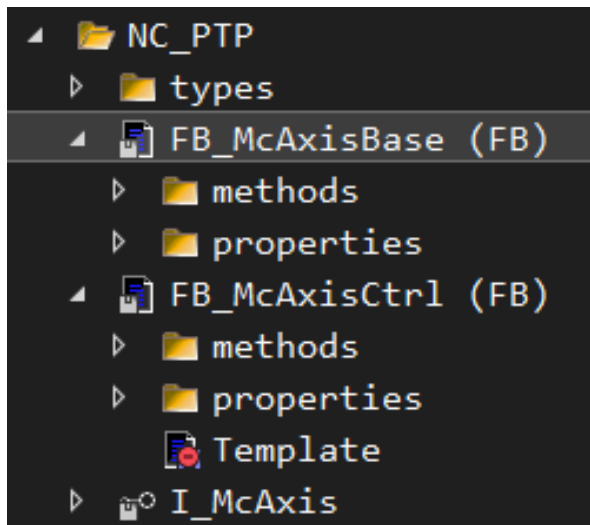
Software Design:

- E_PROGRESS (example log)
 - Different names
 - Same principle

eMessageInfo	2025-10-24-13:00:18.408	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_INIT
eMessageInfo	2025-10-24-13:00:18.418	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_BUSY
eMessageInfo	2025-10-24-13:00:18.458	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_PREPARE
eMessageInfo	2025-10-24-13:00:18.658	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_STARTUP
eMessageInfo	2025-10-24-13:00:18.858	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_CHECK
eMessageInfo	2025-10-24-13:00:18.888	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_OCCUPIED
eMessageInfo	2025-10-24-13:00:18.908	General_ID_0	General	30	ExampleEvalMachine	:TRANSPORT_GROUP_CLEAR:	PROGRESS_DONE

Software Design:

- Class building principle:
 - **Library** wrapper as **ABSTRACT** BaseClass with Interface
 - Ctrl/State wrapper extends BaseClass
 - NC_PTP example

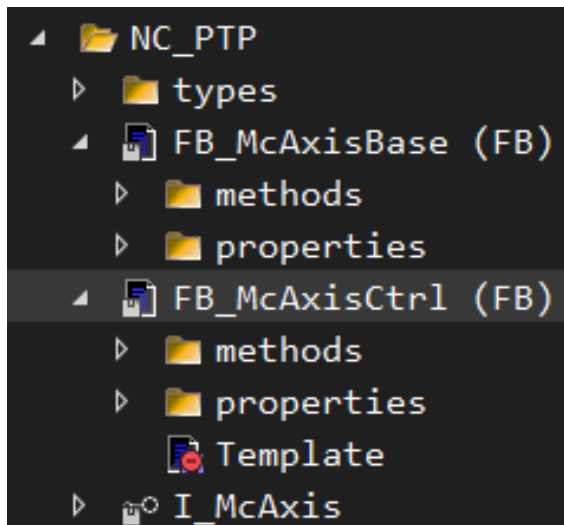


```

FB_McAxisBase
19
20 // last target position datafield
21 _rLastTargetPosition      : LREAL;
22
23 // name and encoder type are read by ADSREAD instance from NC
24 _sAxisName                 : STRING(30) := 'NO NAME'; // AxisName is limited to 30 chars,
25 _sEncoderType              : STRING(30) := 'NO TYPE';
26 _nEncoderType              : UDINT;
27 _eEncoderType              : E_AxisEncoderType;
28
29 // Motion Control FB Instances
30 _fbPower                   : Tc2_MC2.MC_Power;
31 _fbReset                   : Tc2_MC2.MC_Reset;
32 _fbHalt                    : Tc2_MC2.MC_Halt;
33 _fbStop                    : Tc2_MC2.MC_Stop;
34 _fbHome                    : Tc2_MC2.MC_Home;
35 _fbJog                     : Tc2_MC2.MC_Jog;
36 _fbMoveToPosition          : Tc2_MC2.MC_MoveAbsolute;
37 _fbMoveVelo                : Tc2_MC2.MC_MoveVelocity;
38
39 // used for buffered moves
40 _fbMoveVelocity             : ARRAY[1..2] OF Tc2_MC2.MC_MoveVelocity;
41 _fbMoveAbsolute             : ARRAY[1..2] OF Tc2_MC2.MC_MoveAbsolute;
42
43 // ADS read instance for reading startup information from NC
    
```

Software Design:

- Class building principle:
 - Library wrapper as ABSTRACT BaseClass with Interface
 - Ctrl/State wrapper **extends** BaseClass
 - NC_PTP example

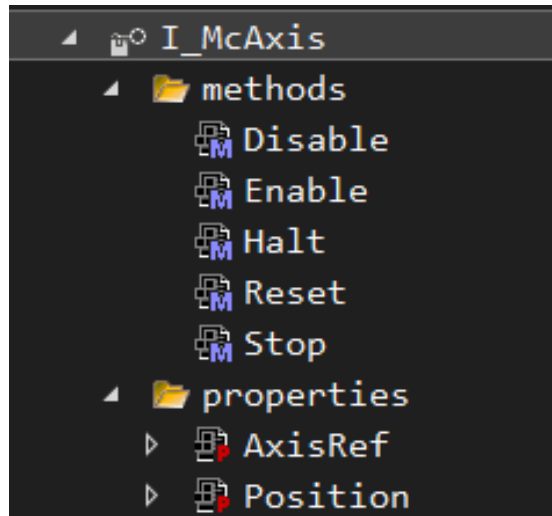


```

FB_McAxisCtrl  FB_McAxisBase
1  FUNCTION_BLOCK FB_McAxisCtrl EXTENDS FB_McAxisBase
2  VAR
3      _eCheck          : E_CHECK_STATE; // pointer check
4
5      _eAxisCycle,
6      _eInit           : E_PROGRESS;    // cyclic AXIS_REF() and MC_Power check
7                                     // axis startup and init, get encoder type,
8
9      _nAxisIndex       : UINT;         // know thyself
10     _rTargetWindow    : LREAL := 0.1; // TODO: get that from NC
11
12     // pointers to all; access required for coupling; access by index
13     _Axes              : REFERENCE TO ARRAY[1..PLC_CONSTANT.MAX_AXIS] OF AXIS_REF;
14
15     _Ctrl              : POINTER TO ARRAY[1..PLC_CONSTANT.MAX_AXIS] OF ST_AXIS_CTRL;
16     _State             : POINTER TO ARRAY[1..PLC_CONSTANT.MAX_AXIS] OF ST_AXIS_STATE;
    
```

Software Design:

- Class building principle:
 - Library wrapper as ABSTRACT BaseClass **with** Interface
- **Interface** for cross-communication between classes
 - NC_PTP example



```
FB_McAxisCtrl  FB_McAxisBase  X
1  FUNCTION_BLOCK ABSTRACT FB_McAxisBase IMPLEMENTS I_McAxis
2  VAR
3      _AxisRef          : REFERENCE TO Tc2_MC2.AXIS_REF;
4      _pCalibrationCam  : POINTER TO BOOL;
5
6      _nErrorId          : UDINT; // datafield for MC error
7
8      _MessageLevel      : E_MessageType;
9
10     // MC_Power inputs for feed fwd, feed bwd, override
11     _bEnablePositive     : BOOL := TRUE;
12     _bEnableNegative     : BOOL := TRUE;
13     _rAxisOverride       : LREAL := 100.0;
14
15     _rValueParameterRead : LREAL;
16     _fbReadParameter      : Tc2_MC2.MC_ReadParameter;
17     _fbWriteParameter     : Tc2_MC2.MC_WriteParameter;
18     _fbWriteParameterPersist : Tc2_MC2.MC_WriteParameterPersistent;
```

Software Design:

PLC_MOTION_LAYER / GVL_AXIS

```
//-----  
// command and state structure  
//-----  
Ctrl  : ARRAY[1..MAX_AXIS] OF  
        ST_AXIS_CTRL;  
  
State : ARRAY[1..MAX_AXIS] OF  
        ST_AXIS_STATE;  
  
//-----  
// cyclic interface function block  
//-----  
Control : ARRAY[1..MAX_AXIS] OF  
        FB_McAxisCtrl;
```

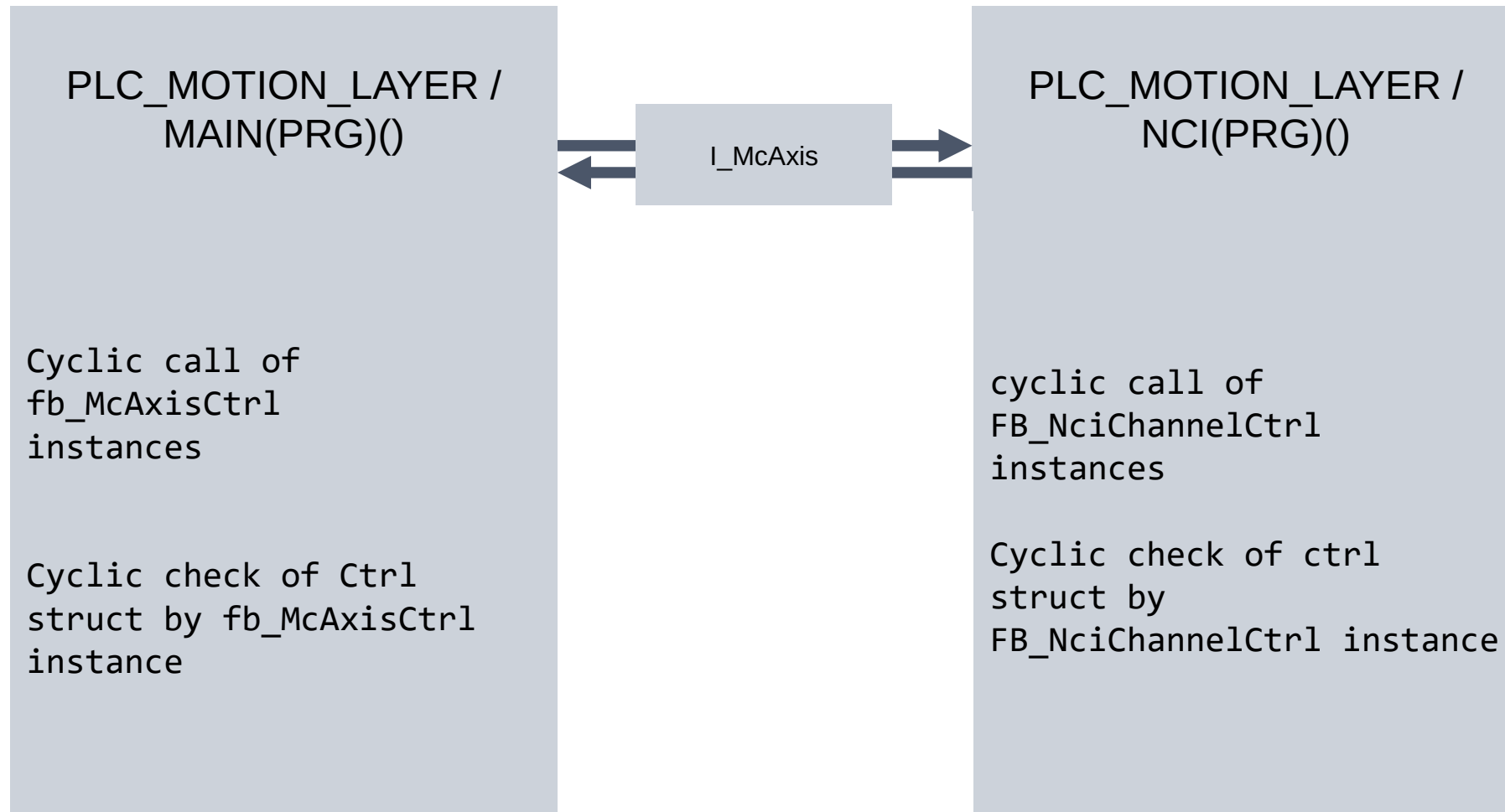
PLC_MOTION_LAYER / GVL_NCI

```
//-----  
// command and state structure  
//-----  
stChannelCtrl  : ARRAY[1..MAX_NCI_CH] OF  
        ST_CTRL_NCI;  
  
stChannelState : ARRAY[1..MAX_NCI_CH] OF  
        ST_STATE_NCI;  
  
//-----  
// cyclic interface function block  
//-----  
fbNciCtrl : ARRAY[1..MAX_NCI_CH] OF  
        FB_NciChannelCtrl;
```

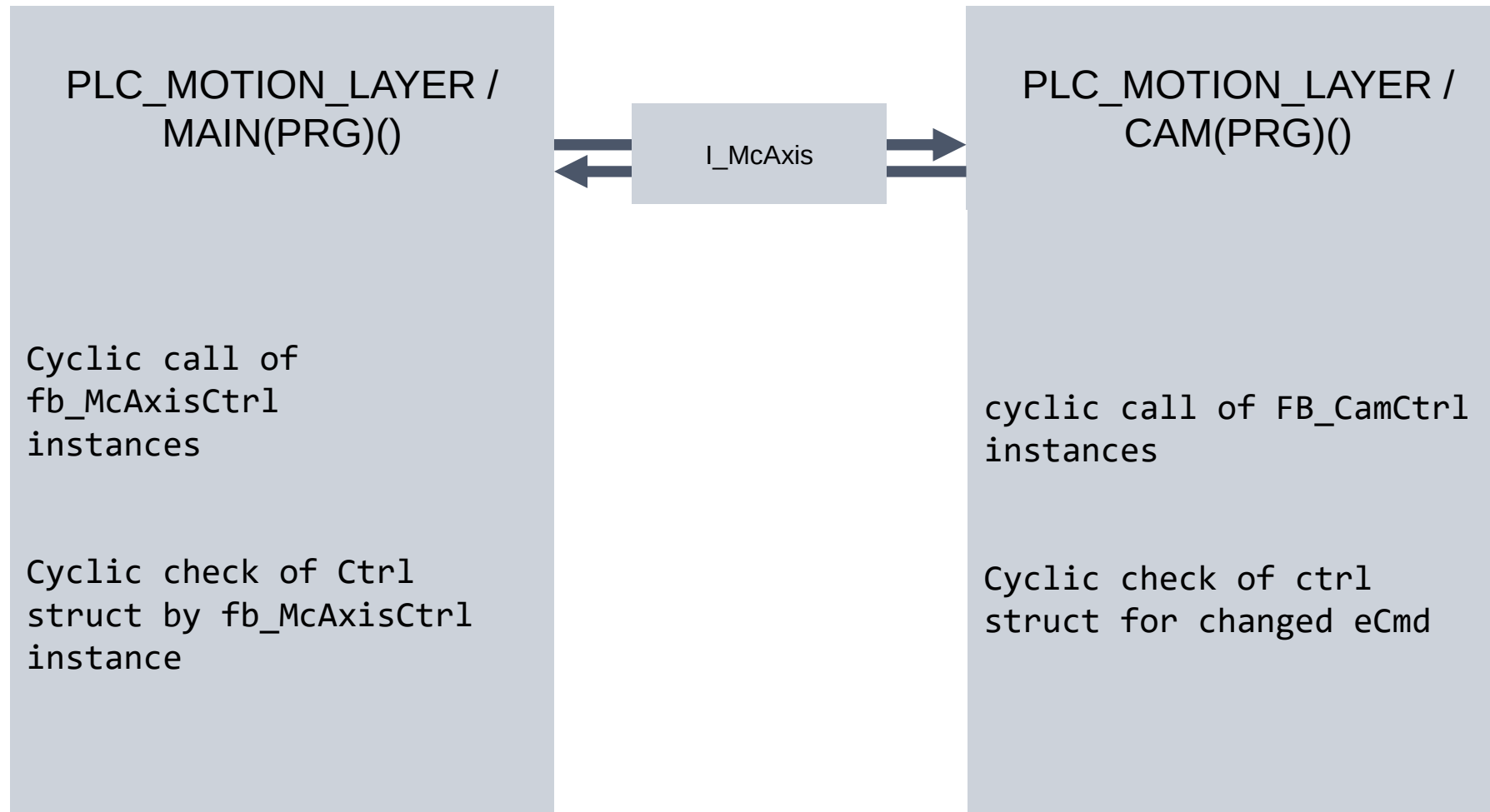
PLC_MOTION_LAYER / GVL_CAM

```
//-----  
// command and state structure  
//-----  
Ctrl  : ARRAY[1..MAX_AXIS] OF  
        ST_CAM_CTRL;  
  
State : ARRAY[1..MAX_AXIS] OF  
        ST_CAM_STATE;  
  
//-----  
// cyclic interface function block  
//-----  
Control : ARRAY[1..MAX_AXIS] OF  
        FB_CamCtrl;
```

Software Design:

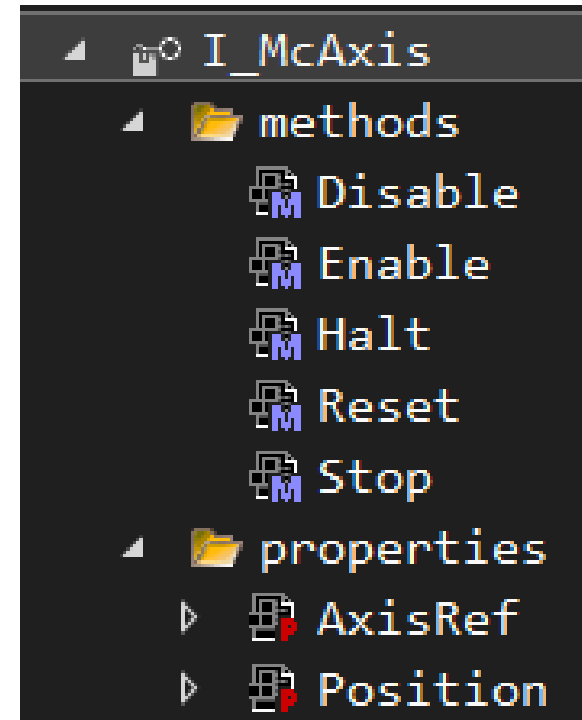


Software Design:



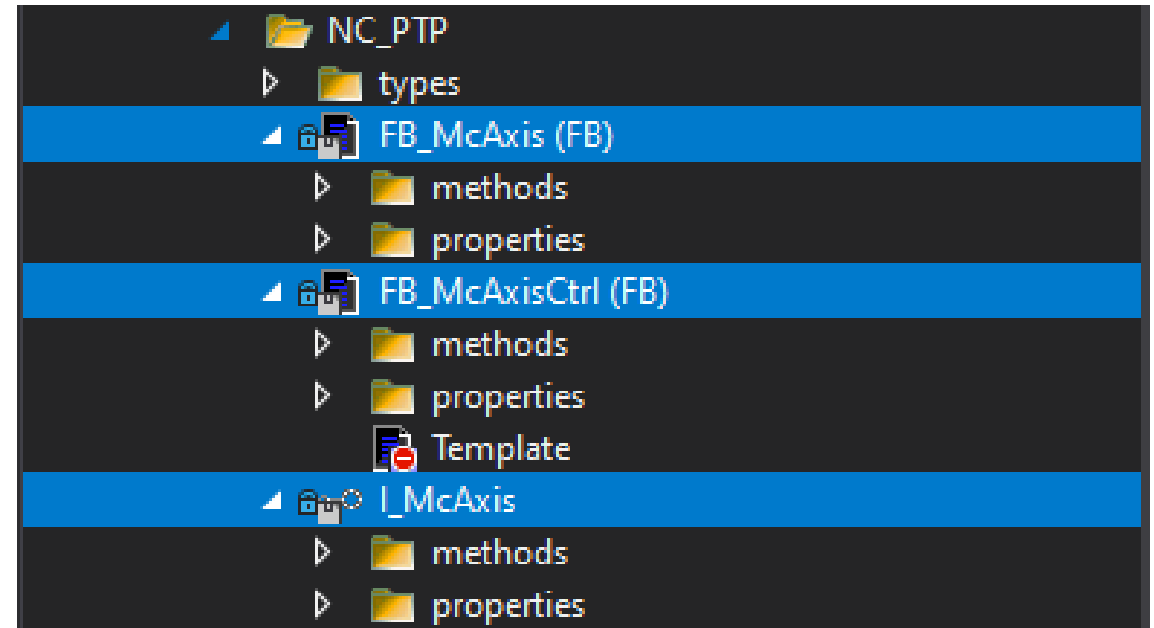
GVL_AXIS:

- I_McAxis: Interface for horizontal access between namespaces
 - Mandatory for using other namespaces
 - Provides essential methods
 - Provides reference to AXIS_REF
 - Direct access to AxisRef.NcToPlc.ActPos



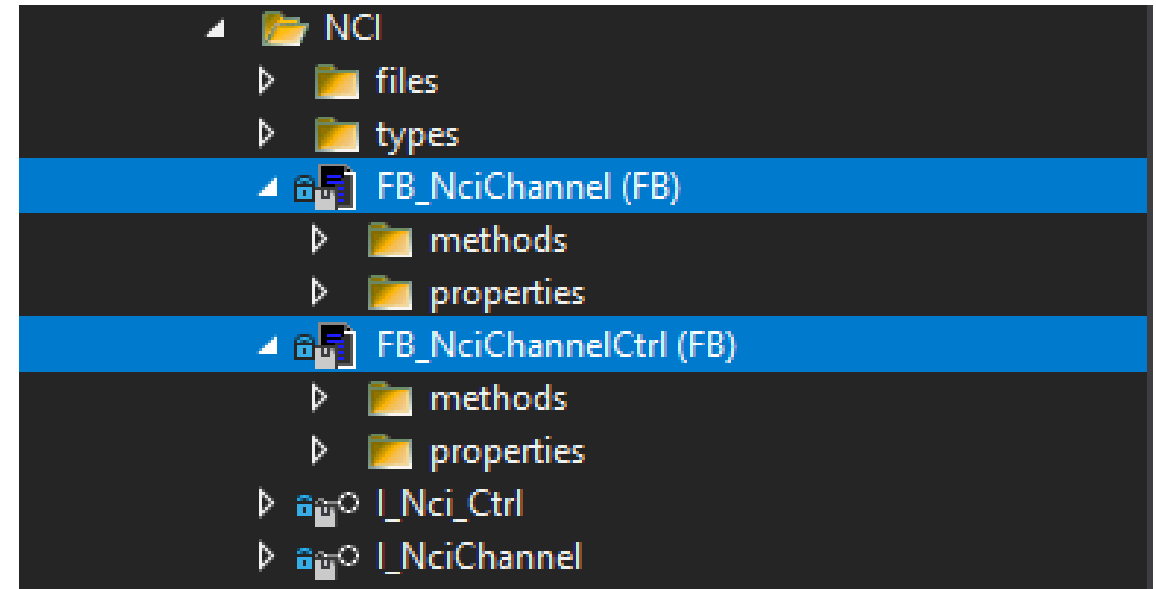
GVL_AXIS:

- McAxis: Point To Point axis
 - Base class **FB_McAxis** wraps Tc2_MC2 function blocks and implements interface
 - **FB_McAxisCtrl** extends base class with cyclic execution wrapper
 - **I_McAxis** is used in advanced motion features (NCI, CAM, XFC)



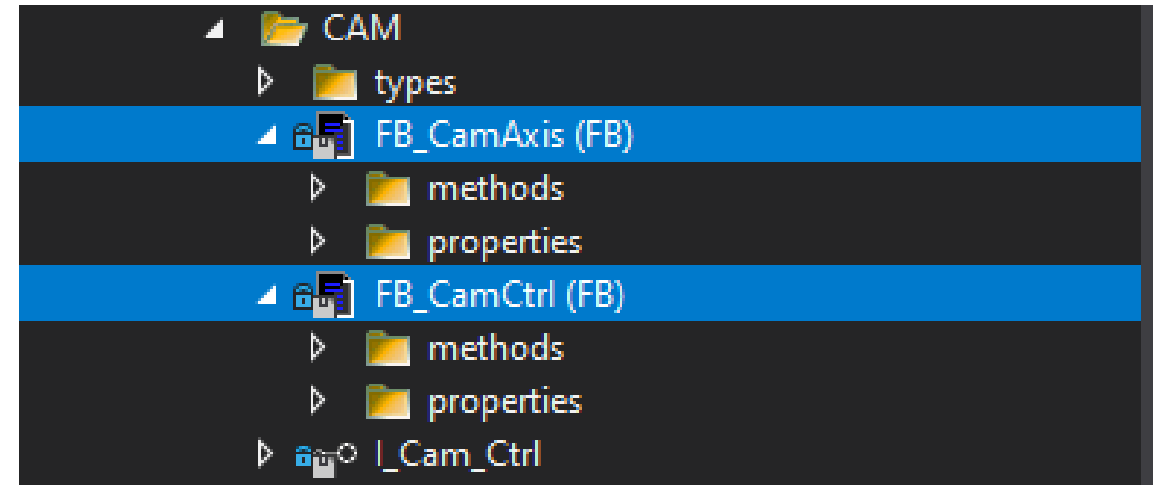
GVL_NCI:

- NCI Channel: XYZ interpolated
 - Base class **FB_NciChannel** wraps Tc2_NCI function blocks and implements interface
- **FB_NciChannelCtrl** extends base class with cyclic execution wrapper and implements interface
- **I_NciChannel** and **I_Nci_Ctrl** are only valid if compiler define is set before compiling project



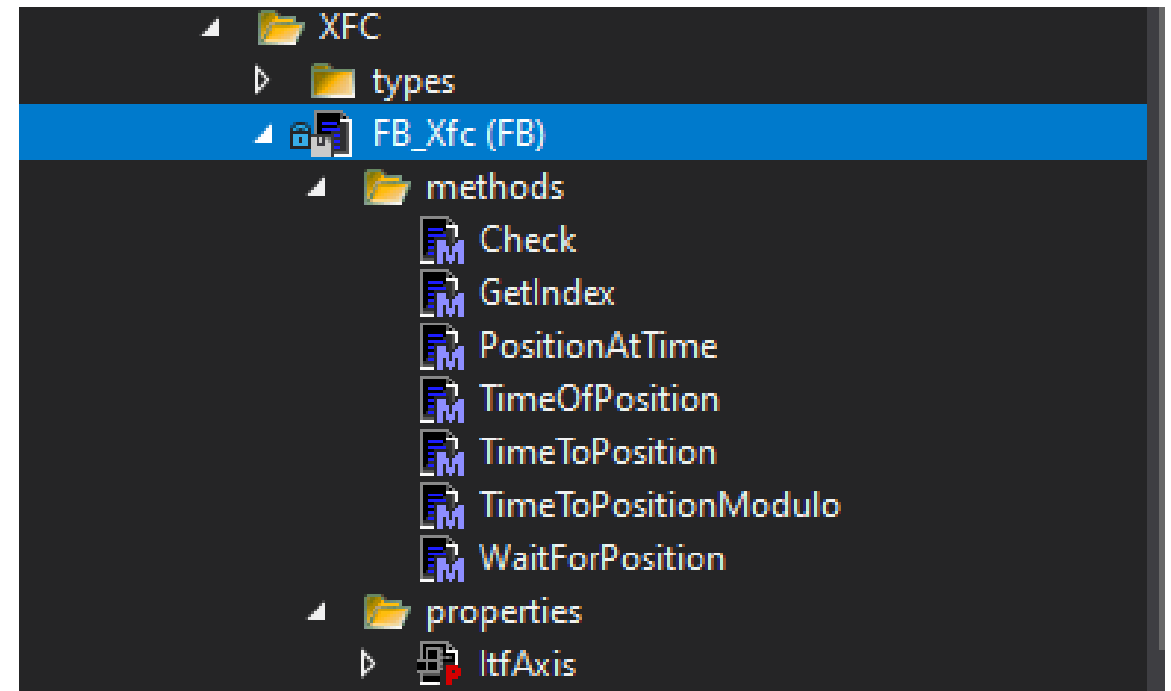
GVL_CAM:

- Camming:
 - Base class **FB_CamAxis** wraps Tc2_MC2_Camming function blocks and implements interface
 - **FB_CamCtrl** extends base class with cyclic execution wrapper and implements interface
 - **I_Cam_Ctrl** is only valid if compiler define is set before compiling project



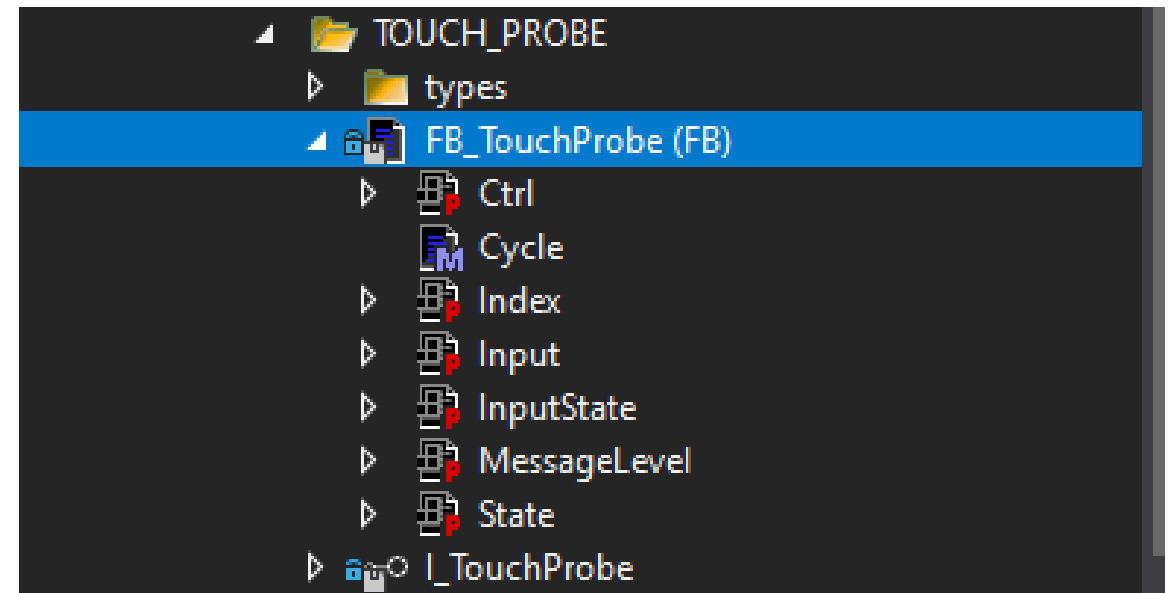
GVL_FUNCTIONS:

- XFC classes:
 - Base class **FB_Xfc** wraps Tc2_MC2_XFC function blocks for Distributed Clock position/time applications



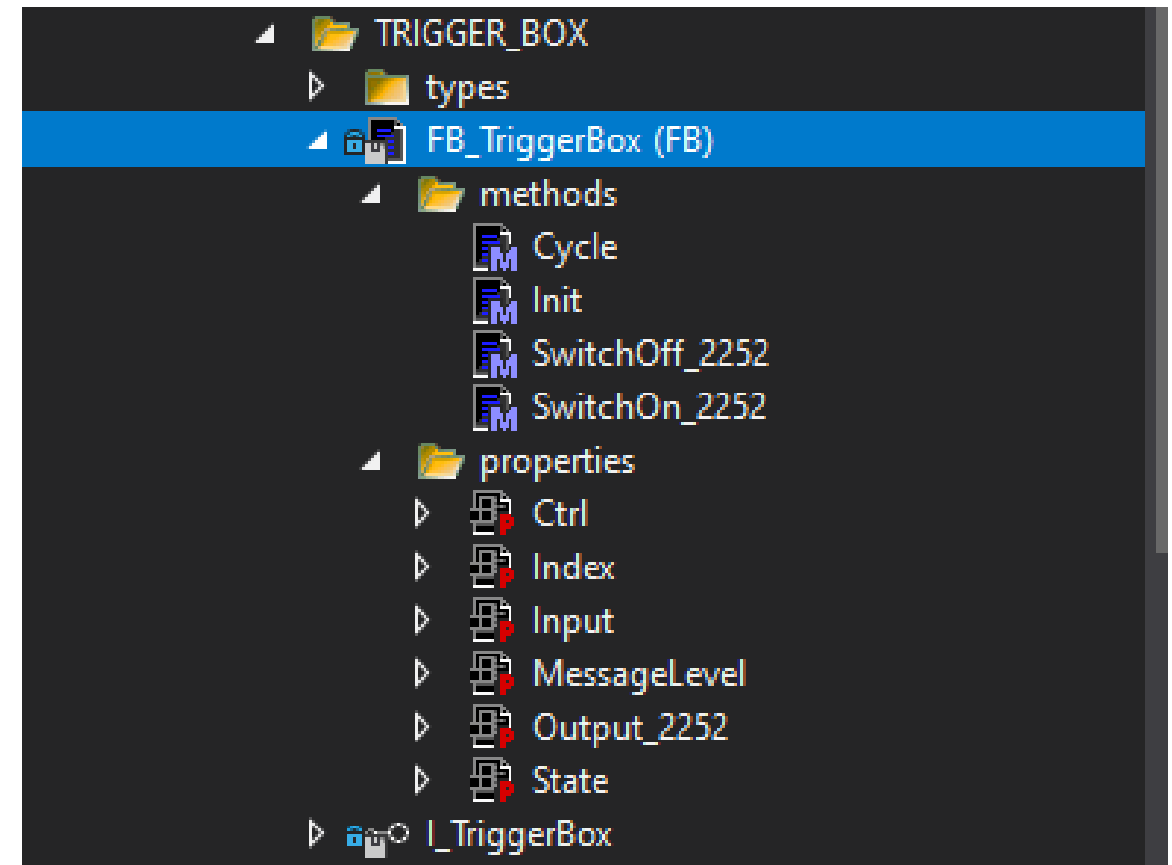
GVL_FUNCTIONS:

- Touch Probe:
 - **FB_TouchProbe** extends **FB_Xfc**
 - Cyclic execution with Ctrl/State pair
 - Must be connected to input device
 - **I_TouchProbe** is only valid if compiler define is set before compiling project



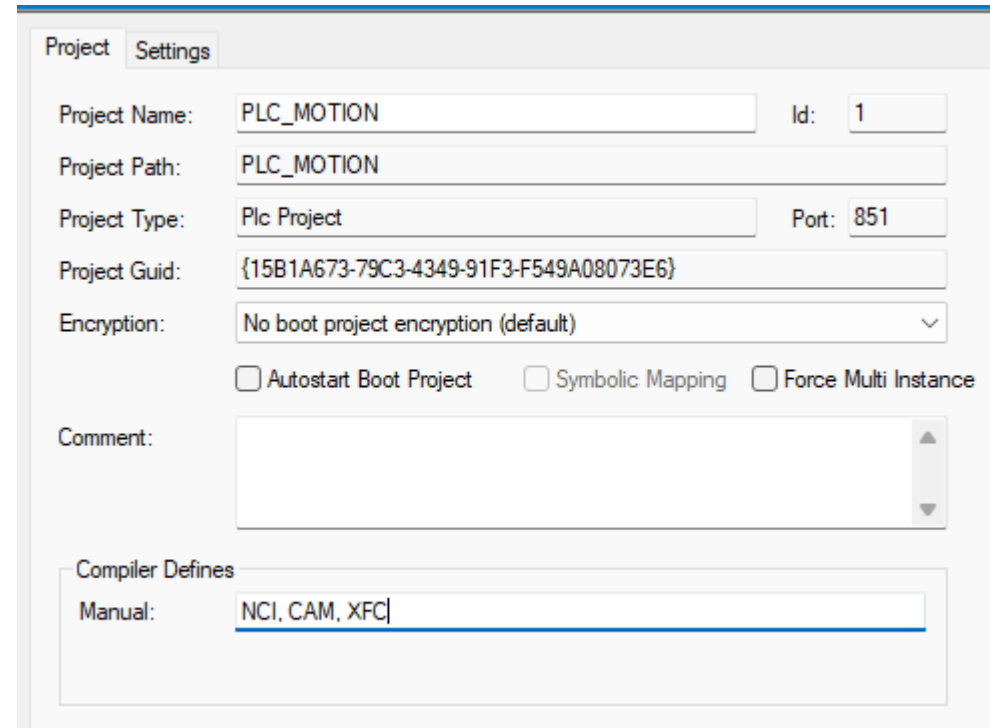
GVL_FUNCTIONS:

- Trigger Box:
 - **FB_TriggerBox** extends **FB_Xfc**
 - Cyclic execution with Ctrl/State pair
 - Must be connected to input/output device
 - **I_TriggerBox** is only valid if compiler define is set before compiling project



Compiler defines / pragmas:

- **BSD**
 - TC-BSD system specific variables are set
- **NCI**
 - NciChannel cyclic interfaces are used
- **CAM**
 - Camming cyclic interfaces are used
- **XFC**
 - TouchProbe and TriggerBox interfaces are used



The screenshot shows the 'Settings' tab of a project configuration window. The 'Project Name' is 'PLC_MOTION' and the 'Id' is '1'. The 'Project Path' is also 'PLC_MOTION'. The 'Project Type' is 'Plc Project' and the 'Port' is '851'. The 'Project Guid' is '{15B1A673-79C3-4349-91F3-F549A08073E6}'. The 'Encryption' is set to 'No boot project encryption (default)'. There are three checkboxes: 'Autostart Boot Project', 'Symbolic Mapping', and 'Force Multi Instance', all of which are currently unchecked. There is a 'Comment' text area which is empty. At the bottom, under 'Compiler Defines', the 'Manual' field contains the text 'NCI, CAM, XFC|'.

Project Name:	PLC_MOTION	Id:	1
Project Path:	PLC_MOTION		
Project Type:	Plc Project	Port:	851
Project Guid:	{15B1A673-79C3-4349-91F3-F549A08073E6}		
Encryption:	No boot project encryption (default)		
☐ Autostart Boot Project ☐ Symbolic Mapping ☐ Force Multi Instance			
Comment:			
Compiler Defines			
Manual:	NCI, CAM, XFC		

Compiler defines / pragmas:

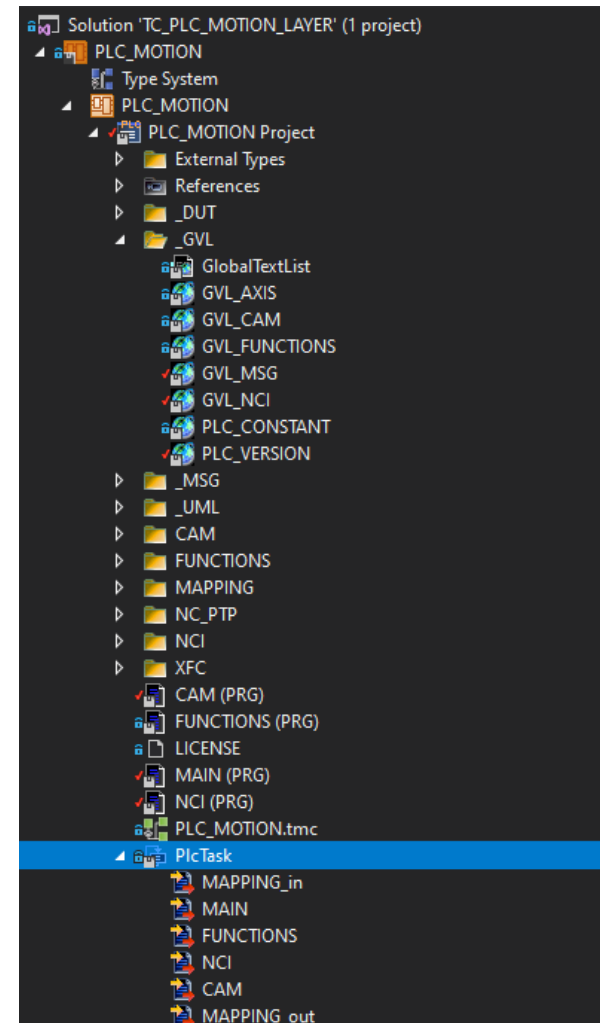
- **AXIS_MAP**
 - MAPPING_in.AxisCtrl is copied onto GVL_AXIS.Ctrl and data structures
 - GVL_AXIS.State is copied onto MAPPING_out.AxisState and info structures

- **CAM_MAP**
 - MAPPING_in.CamControl is copied onto GVL_CAM.Ctrl
 - GVL_CAM.State is copied onto MAPPING_out.CamState and info structures

- **TRIGGER_MAP**
 - TouchProbe and TriggerBox interfaces are used

Cyclic call tree:

- **MAPPING_in(PRG)**
 - Copies mapping data onto input structures
- **MAIN(PRG)**
 - Cyclic call to FB_McAxisCtrl instances
- **FUNCTIONS(PRG)**
 - Cyclic call to TouchProbe and TriggerBox instances
- **NCI(PRG)**
 - Cyclic call to NCI channel instances
- **CAM(PRG)**
 - Cyclic call to FB_CamCtrl instances
- **MAPPING_out**
 - Copies state data onto mapping structures



Logging System:

- Implementation from top down
 - → function based with global timestamp added automatically
- Enumeration based with 4 categories and timestamp
 - → occurrence in strict timestamp order
- Error Id is mirrored directly from called instance
 - → Infosys error numbers can be searched in case of diagnosis
- Optional text for additional information
- Specific logging switch to get more detailed information aside error numbers
- Automated write procedure to ascii formatted file

PLC_MOTION_LAYER project

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